

RESEARCH ARTICLE | OCTOBER 15 2013

Modeling of structural uncertainties in Reynolds-averaged Navier-Stokes closures

Michael Emory; Johan Larsson; Gianluca Iaccarino



Physics of Fluids 25, 110822 (2013)

<https://doi.org/10.1063/1.4824659>



Connected Content

A correction has been published: [Publisher's Note: "Modeling of structural uncertainties in Reynolds-averaged Navier-Stokes closures" \[Phys. Fluids 25, 110822 \(2013\)\]](#)

Articles You May Be Interested In

The deviation from parallel shear flow as an indicator of linear eddy-viscosity model inaccuracy

Physics of Fluids (May 2014)

A framework for epistemic uncertainty quantification of turbulent scalar flux models for Reynolds-averaged Navier-Stokes simulations

Physics of Fluids (May 2013)

Epistemic uncertainty quantification for Reynolds-averaged Navier-Stokes modeling of separated flows over streamlined surfaces

Physics of Fluids (March 2019)



Physics of Fluids

Special Topics Open
for Submissions

[Learn More](#)

Modeling of structural uncertainties in Reynolds-averaged Navier-Stokes closures

Michael Emory,^{1,a)} Johan Larsson,² and Gianluca Iaccarino¹

¹*Department of Mechanical Engineering, Stanford University, Stanford, California 94305, USA*

²*Department of Mechanical Engineering, University of Maryland, College Park, Maryland 20742, USA*

(Received 16 January 2013; accepted 23 September 2013; published online 15 October 2013; publisher error corrected 24 October 2013)

Estimation of the uncertainty in numerical predictions by Reynolds-averaged Navier-Stokes closures is a vital step in building confidence in such predictions. An approach to model-form uncertainty quantification that does not assume the eddy-viscosity hypothesis to be exact is proposed. The methodology for estimation of uncertainty is demonstrated for plane channel flow, for a duct with secondary flows, and for the shock/boundary-layer interaction over a transonic bump. © 2013 AIP Publishing LLC. [<http://dx.doi.org/10.1063/1.4824659>]

I. INTRODUCTION

Reynolds-averaged Navier-Stokes (RANS) turbulence models are widely used in a very broad range of engineering applications. The development of RANS models goes back about three quarters of a century, with increases in mathematical and physical complexity but typically only incremental improvements in accuracy. Despite the success and increased popularity of high-fidelity turbulence simulations (e.g., detached eddy, large eddy, and direct numerical simulations (DNS)) in the academic and research world, RANS models continue to be the primary means of modeling turbulence for engineering applications. The use of computational fluid dynamics (CFD) with RANS models in the engineering design process has grown rapidly over the last three decades. For important design decisions, however, CFD results are almost always complemented with experimental tests and data. Furthermore, the certification of new designs and systems is rarely based on computational results alone.

At the heart of the matter is the issue of how trustworthy a prediction made using a RANS model is for a turbulent flow. This question can have very different answers depending on the situation. For example, one does expect an accurate prediction of the aerodynamic lift, drag, and moments of an airfoil at cruise conditions provided that a suitable RANS model is used. In contrast, one should probably not place much confidence in a RANS prediction of the drag of a bluff body with smooth edges. Specifically, the degree of confidence depends on the model, the flow, and the quantity-of-interest, in addition to other factors. While there clearly exists expert knowledge on these issues in several application fields, it is equally clear that an objective estimation of uncertainty in the quantity-of-interest for a particular flow and/or model is required.

Uncertainty quantification (UQ) of RANS turbulence closures is a fairly new area of research, although efforts in identifying model limitations are common. One of the most basic ideas is to simply compare predictions of different models with respect to some validation dataset. A more formal framework is to consider an ensemble of models and to consider the differences between the predictions of the different models as a measure of the modeling uncertainty; this methodology is employed in weather predictions where multiple computations are used together (as an ensemble) to form the actual prediction.¹ The drawback of this approach is that it may not truly represent the

^{a)}Electronic mail: memory@stanford.edu

full breadth of uncertainty; it is dependent on the models included in the ensemble. For example, an analysis of RANS models might plausibly consist of the Spalart-Allmaras, $k - \varepsilon$, $k - \omega$, shear stress transport (SST), and $v^2 - f$ models.² However, since these models all rely on the eddy-viscosity hypothesis, i.e., a linear (and scalar) stress-strain relationship, the uncertainty introduced by this assumption cannot be captured by examining the ensemble. In other words, statistical comparisons between multiple models can provide insights into the discrepancy but are fundamentally unable to capture structural (or model-form) uncertainties due to modeling assumptions.

A different approach for UQ of RANS models is to assume that the model coefficients (e.g., C_μ , $C_{\varepsilon 1}$, etc.) are uncertain.^{3,4} For example, one might assign ranges or probability distributions for each of the model coefficients and quantify how this affects the quantity-of-interest. Although this approach does characterize the sensitivity of the results to the choice of the coefficients, it completely neglects model-form errors that emerge from inaccuracies in the hypothesized stress-strain relationship as well as errors in the form of the modeled transport equations.

One extension to this approach is the methodology used by Oliver and Moser^{5,6} which combines a data-assimilation procedure for the model coefficients with a stochastic extension to the Boussinesq hypothesis. This approach uses Bayesian statistics and aims at estimating the misfit of the turbulence closure to a set of experimental data. The results obtained are promising although the choice of the stochastic extension and the dependency on relevant experimental data remains a critical component.

The objective of the present work is to formulate a different framework for the estimation of uncertainty in RANS models. Specifically, we seek a framework that: (a) accounts for model-form uncertainty (both the eddy viscosity hypothesis and the form of the modeled PDEs); (b) provides a generalized approach which connects with the physics of turbulence and the relevant assumptions made in the closures; and (c) allows for expert knowledge and the body of experimental validation studies to be part of the uncertainty estimation framework.

We note specifically that the present contribution describes only one piece of the complete framework for uncertainty estimation in RANS models: how the method for uncertainty injection allows for the *shape* of the Reynolds stress tensor to be considered uncertain, and how this creates an uncertainty-framework which, among other things, allows for the consideration of how secondary flows influence the quantity-of-interest. The relationship between the present contribution and the complete framework is described in Sec. II B.

This work stems from the effort in developing a framework for quantification of margins and uncertainties^{7,8} in the prediction of failure modes of a supersonic combustion engine. Although the procedure is derived in the context of turbulence closures, it could be applied to other application areas where tensors are modeled, for example, to constitutive models for elastic and viscoelastic fluids.

The motivation and implementation of this new UQ framework are described in Sec. II together with an analysis of its application to plane shear flow. The framework is demonstrated on several fluid dynamics problems: plane channel flow, square duct flow, and a shock/boundary-layer interaction with flow separation. In Sec. III, we introduce these flows and present results where the spatial dependency of the model form uncertainty is ignored. An informed implementation of the framework which acknowledges and discusses the spatial dependency is presented in Sec. IV, applied again to the same three flows. A summary of the main findings concludes the paper in Sec. V.

II. METHODOLOGY

A. Underlying deterministic RANS

Almost all RANS closures are constructed to estimate the Reynolds stress tensor R_{ij} , defined as $\overline{u'_i u'_j}$ or $\widetilde{u''_i u''_j}$ for incompressible and compressible flow, respectively, with the appropriate use of Reynolds or Favre decompositions $u_i = \bar{u}_i + u'_i = \tilde{u}_i + u''_i$. Most engineering models rely on the eddy-viscosity hypothesis that assumes that the stress R_{ij} is linearly related to the mean rate-of-strain S_{ij} through a scalar eddy-viscosity ν_t , i.e.,

$$R_{ij}^{\text{ev}} = \frac{2k}{3} \delta_{ij} - 2\nu_t S_{ij}^{\text{dev}}, \quad (1)$$

where $k = R_{nn}^{\text{ev}}/2$ is the turbulence kinetic energy and S_{ij}^{dev} is the deviatoric (trace-free) part of the strain-rate tensor $S_{ij} = (\partial_j \tilde{u}_i + \partial_i \tilde{u}_j)/2$ (for variable-density flows $\partial_n \tilde{u}_n \delta_{ij}/3$ should also be subtracted). The superscript “ev” denotes that this assumed stress-strain relationship in Eq. (1) is used only in eddy-viscosity models, including mixing-length models (where k is neglected) as well as one-, two-, and three-equation models, e.g., the Spalart-Allmaras, $k - \omega$, and $v^2 - f$ models.² These models differ in how they compute k and ν_t ; for example, in the commonly used $k - \omega$ model transport equations are solved for k and ω and then $\nu_t = C_\nu k/\omega$. These transport equations include model coefficients that are typically specified by matching known behaviors for canonical flows, e.g., the decay-rate in isotropic turbulence or the log-law in turbulent channel flow.²

B. Overview: The general framework

The present contribution is a part of a general framework and philosophical approach to uncertainty estimation of RANS models. Note the deliberate use of the word “estimation,” rather than “quantification”: RANS models are sophisticated and mathematically complex with nonlinearity and strong coupling to the velocity field. This implies that the evaluation of provable bounds on the quantity-of-interest (i.e., a formal confidence interval) will most likely not be tight, in fact much larger than what any experienced practitioner would deem reasonable. Therefore, our aim here is *not* on developing rigorous provable bounds, but rather on producing a reasonable and informed estimate of the uncertainty in the RANS predictions.

The possible uncertainty in R_{ij} is clearly greater than the values of the model coefficients in a postulated combination of stress-strain relationship and transport equations. One key objective of the proposed framework is to include such uncertainties, i.e., the framework should be able to address the full range of possible uncertainties.

Countering this, however, is the fact that there exists (in the community of turbulence modelers) an enormous amount of validation studies and experience that fosters confidence in the predictive power of RANS models, at least for certain types of flows. For example, a properly executed RANS prediction of the lift of an airfoil at cruise is a reliable and trustworthy prediction; a RANS prediction of the stall angle is not. Thus, the uncertainty framework should also not be overly conservative, and should allow for validation experiences and expert knowledge to influence (and often decrease) the estimated uncertainty.

The apparent contradiction between the last two aspects is resolved in the following manner to create the complete uncertainty framework: (1) Given a RANS prediction, the flow field is divided into regions where we trust the model and regions where we do not; this is accomplished by a “marker function.” (2) Since the model is considered uncertain only in the marked regions, perturbations to the RANS model are injected only there. (3) Solution of the full equations then propagates this uncertainty to the final quantity-of-interest.

The present contribution covers primarily the second item (and the third, which is trivial) where the objective is to develop the uncertainty-injection methodology. Specifically, we focus on injecting uncertainty in the shape (or anisotropy) of the Reynolds stress tensor and quantify if there is any fundamental bias induced in the predictions. The approach of injecting uncertainty directly into properties of the Reynolds stress has the added benefit of treating the underlying RANS model implementation as a black box, merely outputting the Reynolds stress as a function of the velocity field $\tilde{u}_i$.

Although the perturbations considered in this work are injected only in the anisotropy eigenvalues, the coupled nature of the governing equations directly influences the mean flow and the turbulent kinetic energy; consequently, the gradients of the mean velocity and, therefore the turbulence anisotropy eigenvectors, are altered with respect to the baseline model, albeit not directly. The focus on quantifying uncertainty in the Reynolds stress eigenvalues has deep significance in turbulence applications as it is well known that normal stress imbalances can determine secondary motions in corner flows and wall jets⁹ or affect the decay of swirling motions in a combustion chamber;¹⁰ the methodology illustrated here estimates the potential effect of these phenomena on the overall flow development without introducing additional modeling assumptions or closures.

Moreover, the study of homogeneous anisotropic flows¹¹ continues to be a critical building block toward the construction of more advanced models and the present effort provides an insight into the effect of introducing different (although realizable) anisotropy representations in more complex flows. Results of uncertainty estimation utilizing a fundamentally different implementation of this framework where both kinetic energy and turbulence anisotropy are perturbed is discussed in Gorlé and Iaccarino.¹²

The issue of the “marker function” and how to decide which regions in space are those where the model is plausibly erroneous (or at least not very trustworthy) is a difficult yet essential part of the proposed framework. In this work, the spatially varying nature of the perturbations is discussed and a variety of possible marker functions are demonstrated. Rigorous discussion and validation of a generically defined marker is left for future contributions (some discussion of this is given in Gorlé and Iaccarino¹²).

C. R_{ij} decomposition and the barycentric map

The focus of the present work is to model the uncertainty in the predicted R_{ij} , in a way agnostic to the RANS model used. The RANS model implementation is viewed as a black box; among other things, this implies that the uncertainty injection machinery presented here can be applied to non-eddy-viscosity models as well. To this end, it is useful to decompose the modeled Reynolds stress into factors determining its amplitude, shape, and orientation.¹³ This is done by introducing the turbulence anisotropy tensor a_{ij} as

$$a_{ij} = \frac{R_{ij}}{2k} - \frac{1}{3}\delta_{ij}. \quad (2)$$

Ensuring physically plausible values for the Reynolds stress – the condition of realizability¹⁴ – requires that the eigenvalues of R_{ij} are non-negative and that the Cauchy-Schwartz inequality holds for the off-diagonal components. The constraints on the individual components of R_{ij} along with Eq. (2) specify that $a_{ij} \in [-1/3, 2/3]$ for $i = j$ and that $a_{ij} \in [-1/2, 1/2]$ for $i \neq j$. Furthermore, an eigen-decomposition of the (symmetric trace-free) anisotropy tensor yields

$$a_{in}v_{nl} = v_{in}\Lambda_{nl}, \quad (3)$$

where v_{ij} is the matrix of orthonormal eigenvectors and Λ_{nl} is the traceless diagonal matrix of eigenvalues λ_l . Multiplication by v_{jl} yields $a_{ij} = v_{in}\Lambda_{nl}v_{jl}$ and thus the Reynolds stress can be decomposed as

$$R_{ij} = 2k \left(\frac{1}{3}\delta_{ij} + v_{in}\Lambda_{nl}v_{jl} \right). \quad (4)$$

The tensors v and Λ are ordered such that $\lambda_1 \geq \lambda_2 \geq \lambda_3$. Note that the amplitude, shape, and orientation of R_{ij} in Eq. (4) are directly represented by turbulence kinetic energy k , and the turbulence anisotropy eigenvalues λ_l and eigenvectors v_{ij} , respectively.

The eigen-decomposition of a_{ij} is useful to separate the shape and the orientation of the Reynolds stress tensor from each other, but also to directly enforce the realizability constraints on R_{ij} through the barycentric map.¹⁵ Consider an equilateral triangle in two dimensions defined by the three corners $\mathbf{x}_{1c}$, $\mathbf{x}_{2c}$, and $\mathbf{x}_{3c}$, where $\mathbf{x} = (x, y)$ is the coordinate in this space – Fig. 1(a). Then

$$\mathbf{x} = \mathbf{x}_{1c}(\lambda_1 - \lambda_2) + \mathbf{x}_{2c}(2\lambda_2 - 2\lambda_3) + \mathbf{x}_{3c}(3\lambda_3 + 1), \quad (5)$$

defines a linear mapping between the anisotropy eigenvalues and the coordinates $\mathbf{x}$. This mapping, together with the requirement that the sum of the eigenvalues is zero, is an invertible one-to-one mapping that can be written as, say, $\mathbf{x} = B\lambda_l$. The realizability constraint implies that $\mathbf{x}$ must lie within the triangle, which is simple to visualize and enforce.

Additionally, it is important to note that the corners of the barycentric map represent different limiting physical behaviors of turbulence, and any realizable state of turbulence is a convex combination of these limiting behaviors – see Eq. (5).

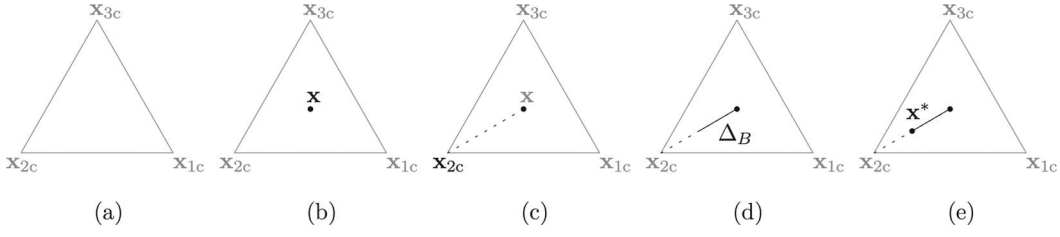


FIG. 1. Progression depicting the eigenvalue sampling procedure within the barycentric map (a). The λ_l from the base-model produces an initial location $\mathbf{x}$ within the map (b). A line is drawn (c) from $\mathbf{x}$ to $\mathbf{x}^{(t)}$ ($\mathbf{x}_{2c}$ in this example), and a relative distance $\Delta_B \in [0, 1]$ along this line (d) is used to define the location $\mathbf{x}^*$ (e), which uniquely maps to λ_l^* .

D. Perturbed R_{ij}^*

To model errors in the underlying turbulence closure, uncertainty is injected into the decomposed Reynolds stress defined in Eq. (4). We define the perturbed Reynolds stress as

$$R_{ij}^* = 2k^* \left(\frac{1}{3} \delta_{ij} + v_{in}^* \Lambda_{nl}^* v_{jl}^* \right), \quad (6)$$

where $k^* = k + \Delta k$ is the perturbed turbulence kinetic energy and Λ_{kl}^* is the diagonal matrix of perturbed eigenvalues λ_l^* . The perturbed eigenvector matrix is defined as $v_{ij}^* = q_{in} v_{nj}$ where q_{in} is an orthonormal rotation matrix. Note that there are 6 degrees-of-freedom in these perturbations: one for the amplitude, two for the shape (since the sum of the eigenvalues is zero), and three for the orientation (since the eigenvectors are orthonormal).

Perturbing k is an important component of the general framework, since the full range of uncertainties will rarely be captured without perturbations to k . While important, perturbing k is rather straightforward, and does not generate surprising results; therefore, it is not covered in this paper. In general, one should introduce perturbations to the orientation matrix as well; this is considered in Gorlé and Iaccarino.¹² For these reasons, we restrict the current contribution to investigating perturbations to Λ^* . The perturbed eigenvalues are defined implicitly through the coordinates in the barycentric map. In other words, $\lambda_l^* = B^{-1} \mathbf{x}^*$ with the perturbation defined in the barycentric map.

For the eigenvalues determining the shape of the Reynolds stress tensor, the perturbation has both a magnitude (the distance $|\mathbf{x}^* - \mathbf{x}|$) and a direction within the barycentric map, both of which could vary in space. The issue of how to specify the spatial variation of these perturbation parameters (amplitudes, directions, etc.) is addressed in Sec. IV. We consider perturbations in the barycentric map in only three directions: toward each of the three corners of the triangle $\mathbf{x}_{1c}$, $\mathbf{x}_{2c}$, and $\mathbf{x}_{3c}$. The perturbations are defined through a relative distance Δ_B of movement toward the target corner. Thus, the perturbed location in the barycentric map is

$$\mathbf{x}^* = \mathbf{x} + \Delta_B (\mathbf{x}^{(t)} - \mathbf{x}), \quad (7)$$

where superscript “(t)” denotes the target corner and $\mathbf{x}$ is the base-model prediction. This process illustrated in Fig. 1 where $\mathbf{x}^{(t)} = \mathbf{x}_{2c}$ and $\Delta_B = 0.5$.

With the linear map B , we describe the perturbed eigenvalues

$$\lambda_l^* = B^{-1} \mathbf{x}^*, \quad (8a)$$

$$= (1 - \Delta_B) B^{-1} \mathbf{x} + \Delta_B B^{-1} \mathbf{x}^{(t)}, \quad (8b)$$

$$= (1 - \Delta_B) \lambda_l + \Delta_B B^{-1} \mathbf{x}^{(t)}. \quad (8c)$$

Taking the three corners $\mathbf{x}_{1c}$, $\mathbf{x}_{2c}$, and $\mathbf{x}_{3c}$ as the target states, we have $B^{-1}\mathbf{x}_{1c} = (2/3, -1/3, -1/3)^T$, $B^{-1}\mathbf{x}_{2c} = (1/6, 1/6, -1/3)^T$, and $B^{-1}\mathbf{x}_{3c} = (0, 0, 0)^T$.

These corners represent limiting states of one-, two-, and three-component (isotropic) turbulence, where $\mathbf{x}_{2c}$ describes specifically the limiting case of two-component turbulence where $\lambda_1 = \lambda_2$. The componentiality of turbulence refers to the number of non-zero Reynolds stress eigenvalues, or alternatively the number of $\lambda_i \neq -1/3$. One major benefit of perturbing toward these limiting states is that the results of the perturbations become easier to interpret in a physically meaningful manner. For example, it is well known that turbulence approaches a two-component state close to a solid wall; results of this perturbation can then be interpreted in terms of whether the model respects this limiting behavior or not. The fact that the perturbed stress satisfies the realizability constraint (provided that the base-model does) is a second benefit.

E. Example: Plane shear flow

Consider a locally plane shear flow where $\partial_2 \tilde{u}_1 = S$ is the only non-zero component of the velocity gradient tensor. An eddy-viscosity RANS model would then generate the anisotropy tensor $a_{12}^{ev} = a_{21}^{ev} = -v_t S/k$ with all other components equal to zero. The eigenvectors for this case would be directed $\pm 45^\circ$ from the horizontal axis, and along the span-wise direction.

If a more general (non-eddy-viscosity model) is used, then we only know that $a_{13} = a_{23} = a_{31} = a_{32} = 0$. For this case, it is easy to show that $v_{i2} = (0, 0, 1)^T$ is still an eigenvector, and that the remaining two eigenvectors, associated with the largest and smallest eigenvalues, respectively, are $v_{i1} = (\cos \theta, -\sin \theta, 0)^T$ and $v_{i3} = (\sin \theta, \cos \theta, 0)^T$, where θ is an angle from the horizontal axis.

For a plane shear flow, only the R_{12} component of the Reynolds stress has a direct effect on the mean momentum equation. From Eq. (6), we have

$$a_{12} = (\lambda_3 - \lambda_1) \sin(2\theta)/2. \quad (9)$$

Therefore, perturbation of the intermediate eigenvalue (more precisely, the eigenvalue associated with the eigenvector normal to the plane of strain) has at most an indirect effect on the mean momentum transport, while perturbations of the largest and smallest eigenvalues are of equal importance. We visualize the magnitude of $\lambda_3 - \lambda_1$ for all locations within the barycentric map (Fig. 2), which allows for quick determination of the local change in shear magnitude for different choices of $\mathbf{x}^{(t)}$ and Δ_B .

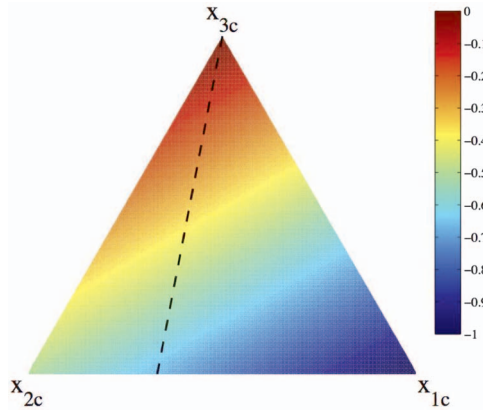


FIG. 2. Barycentric map filled by contours of $\lambda_3 - \lambda_1$, which is related to the shear stress in plane shear flow. The anisotropy becomes zero at $\mathbf{x}_{3c}$ while $|a_{12}|$ is maximized at $\mathbf{x}_{1c}$. Plane shear flow, where at least one $\lambda_i = 0$, is denoted by the dashed black line.

Of greater interest is understanding the effect of turbulence anisotropy from the shape perturbation

$$\Delta a_{12}|_{\mathbf{x}_{1c}} = -\Delta_B \frac{1}{2} \sin 2\theta (\lambda_3 - \lambda_1 + 1) = -\Delta_B \left(a_{12} + \frac{1}{2} \sin 2\theta \right), \quad (10a)$$

$$\Delta a_{12}|_{\mathbf{x}_{2c}} = -\Delta_B \frac{1}{2} \sin 2\theta \left(\lambda_3 - \lambda_1 + \frac{2}{3} \right) = -\Delta_B \left(a_{12} + \frac{1}{3} \sin 2\theta \right), \quad (10b)$$

$$\Delta a_{12}|_{\mathbf{x}_{3c}} = -\Delta_B \frac{1}{2} \sin 2\theta (\lambda_3 - \lambda_1) = -\Delta_B a_{12}, \quad (10c)$$

where $\Delta a_{12}|_{\mathbf{x}^{(i)}}$ is the change in a_{12} relative to the base-model prediction when perturbing toward a specific target state. For perturbations toward $\mathbf{x}_{3c}$, a larger Δ_B causes the value $|a_{12}|$ to decrease and for $\Delta_B = 1$ the magnitude (and thus the shear stress) becomes zero. Moving toward the one- and two-component corners, we observe that as $\Delta_B \rightarrow 1$ a finite value for a_{12}^* is reached: either $(-\sin 2\theta)/2$ or $(-\sin 2\theta)/3$, for $\mathbf{x}_{1c}$ and $\mathbf{x}_{2c}$ perturbations, respectively. When the base-model prediction is between zero and $(-\sin 2\theta)/3$, then perturbations toward $\mathbf{x}_{2c}$ and $\mathbf{x}_{1c}$ will both increase $|a_{12}|$, the difference being $\Delta a_{12}|_{\mathbf{x}_{1c}} > \Delta a_{12}|_{\mathbf{x}_{2c}}$, confirmed visually in Fig. 2.

III. RESULTS: UNIFORM Δ_B

The proposed strategy for uncertainty injection into the Reynolds stress is demonstrated on a sequence of problems including incompressible channel flow and square duct flow, and the transonic flow over a bump. Results corresponding to the SST¹⁶ model without any perturbations are referred to hereafter as base-model solutions. Simulations where uncertainty has been injected are described by their respective target corner ($\mathbf{x}_{1c}$, $\mathbf{x}_{2c}$, or $\mathbf{x}_{3c}$) and the perturbation distance (Δ_B) within the barycentric map.

Only perturbations to the anisotropy of the Reynolds stress tensor are discussed here, since this is the novel aspect of the present framework. Results where both the anisotropy and the turbulent kinetic energy were perturbed were shown by Gorié and Iaccarino.¹² In this section, the perturbation magnitude parameter Δ_B is assumed uniform in space; this assumption is relaxed in Sec. IV.

The JOE code is used to solve the compressible RANS equations. The code uses an unstructured finite-volume discretization that is nominally second order accurate in space, together with the HLLC approximate Riemann solver and slope limiters to ensure robust solutions around shock waves. The code is described by Pečnik *et al.*¹⁷

A. Plane channel flow

We first evaluate a canonical channel flow at $Re_\tau = 2003$ for which DNS results are available.¹⁸ The computations are performed at a Mach number of 0.2 with fixed mass flux, which implies that the friction Reynolds number Re_τ differs slightly for the perturbed cases. The mesh has $y^+ \approx 0.1$ for the first cell, with 80 cells up to the symmetry line.

The results obtained with perturbed anisotropy (i.e., perturbed Λ^*) toward all three limiting states using $\Delta_B = 0.5$ are presented in Fig. 3. Profiles of both mean velocity and shear stress clearly show an “enveloping” behavior with respect to the base-model prediction, in the sense that perturbations toward isotropy ($\mathbf{x}_{3c}$) decreases the shear stress (thus increases u^+) while perturbations toward the other two corners do the opposite. This can be explained by Eq. (10), which shows that perturbations toward $\mathbf{x}_{1c}$ increases the stress more than perturbations toward $\mathbf{x}_{2c}$ (recall that a_{12} is negative), and also that perturbations toward both of these target states increase the stress provided that $a_{12} < 1/3$ (true throughout the channel; also note that $\theta = 45^\circ$ for this eddy-viscosity model).

The effect of changing Δ_B , the magnitude of the perturbation, is shown in Fig. 4. As expected, a larger Δ_B leads to larger variation in the results. Note that $\Delta_B = 1$ for the $\mathbf{x}_{1c}$ perturbation completely removes the Reynolds stress, thus producing a laminar solution.

It is important to note that the perturbed solutions only envelope the *predicted* base-model result, not necessarily the true solution. For example, the DNS results in the buffer layer ($5 < y^+ < 50$) of

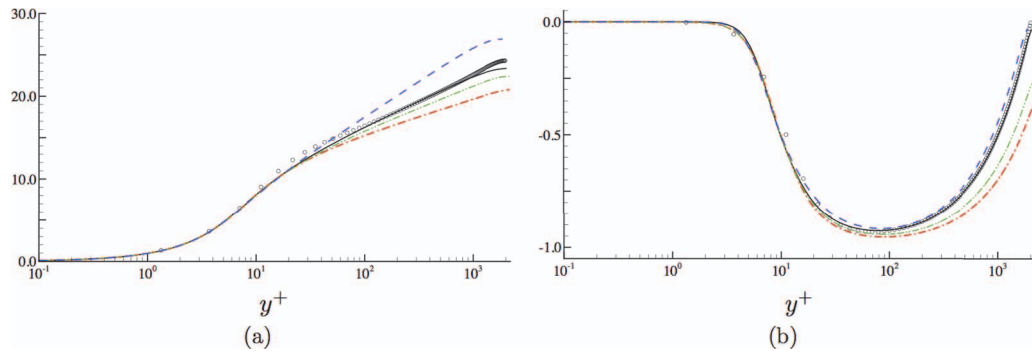


FIG. 3. Channel flow profiles of u^+ (a), and $\overline{u'v'}/u_\tau^2$ (b) versus distance from the wall. In both quantities, the base-model prediction (solid line) is *enveloped* by the eigenvalue perturbations with $\Delta_B = 0.5$. The $\mathbf{x}_{3c}$ perturbation (dashed) over-predicts the base-model while both $\mathbf{x}_{2c}$ (dashed-dotted-dotted) and $\mathbf{x}_{1c}$ (dashed-dotted) perturbations under-predict the base-model. DNS data¹⁸ (circles – every 3rd point shown) is displayed for reference.

Fig. 3(a) are not strictly within the envelope; to accomplish this requires perturbations to k in this region.

B. Square duct flow

The flow in a square duct of cross-section $2h \times 2h$ is investigated at $Re_\tau = 2003$. A similar grid as in Sec. III A is used, where 80×80 cells cover a quarter of the duct. A key feature of duct flows is the presence of Prandtl's second kind of secondary flow, where the anisotropy of the turbulent stress generates mean stream-wise vorticity in the form of corner vortices.^{9,19} It is well known that eddy-viscosity models are incapable of producing this type of secondary flow, since they predict equal normal stresses whenever $\partial_x U = \partial_y V = \partial_z W = 0$. The secondary flow is visualized in Fig. 5 for the $\mathbf{x}_{1c}$ and $\mathbf{x}_{2c}$ perturbations; the $\mathbf{x}_{3c}$ perturbation maintains the isotropy of the normal stresses, and thus fails to produce any secondary flow.

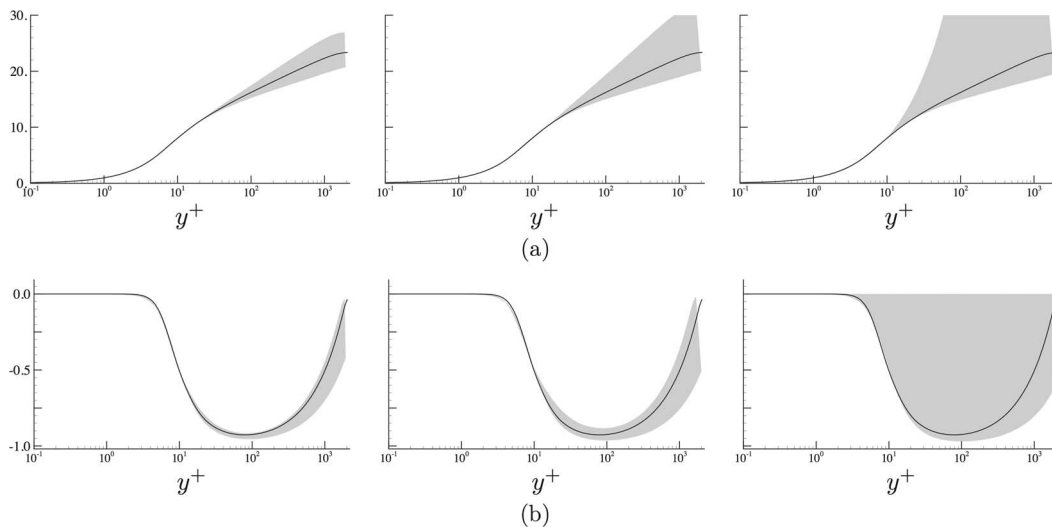


FIG. 4. Channel flow profiles of u^+ (a) and $\overline{u'v'}/u_\tau^2$ (b) versus distance from the wall showing the envelopes (gray region between $\mathbf{x}_{3c}$ and $\mathbf{x}_{1c}$ profiles) for eigenvalue perturbations with different values of Δ_B . From left to right are $\Delta_B = 0.5$, 0.75 , and 1.0 perturbation results, displaying a clear relationship between the amplitude of perturbation and the size of the envelope. The base-model solution (solid line) is provided for reference.

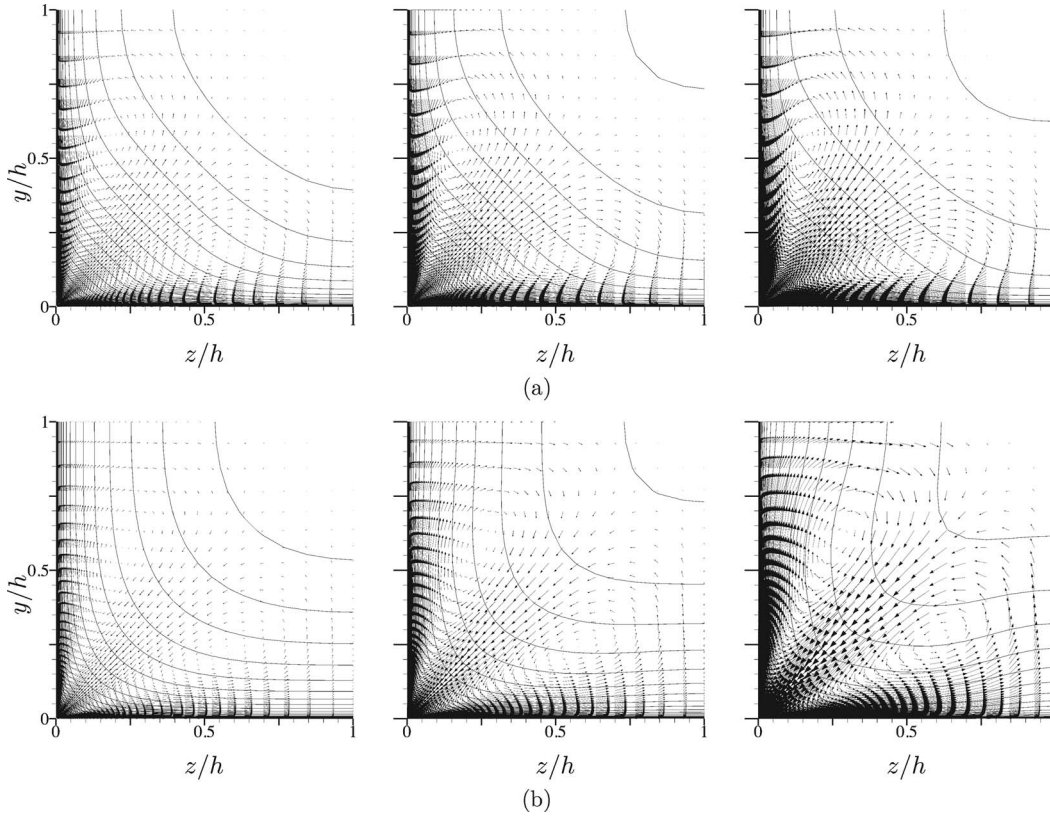


FIG. 5. Secondary flow in a square duct for $\mathbf{x}_{1c}$ (a) and $\mathbf{x}_{2c}$ (b) eigenvalue perturbations. Contour-lines of the stream-wise velocity and in-plane velocity vectors are used to visualize the secondary flows. In both (a) and (b), secondary flows are generated, but with different directions. The strength of the secondary flows are related to the amplitude of the perturbation. From left to right are results with $\Delta_B = 0.25, 0.5$, and 1.0 .

Fig. 5 shows two things: first, the magnitude of the secondary flows are directly related to Δ_B , the magnitude of the perturbation. Second, and more interestingly, the direction of the secondary flow is correlated to the target state $\mathbf{x}^{(1)}$. Specifically, the perturbation toward the $\mathbf{x}_{2c}$ corner produces the correct behavior of flow impinging toward the corner. On the other hand, perturbation toward the $\mathbf{x}_{1c}$ corner produces the opposite, and completely unphysical, flow.

It is easy to show that the production term in the stream-wise vorticity equation is directly proportional to $\overline{v^2} - \overline{w^2}$, i.e., the difference of the normal stresses in the duct cross-section.¹⁹ The perturbations to the anisotropy eigenvalues effectively redistribute the magnitude of normal stress among the three principal component directions. The base-model predicts equal-valued normal stresses along the inertial coordinates, thus $\overline{v^2} = \overline{w^2}$; perturbing toward $\mathbf{x}_{3c}$ only maintains this equality. Moving toward the other limiting states, however, causes a stress imbalance and induces stream-wise vorticity production.

Profiles of $\overline{v^2} - \overline{w^2}$ at different duct span-wise locations are reported in Fig. 6 for perturbations with $\Delta_B = 0.5$. The $\mathbf{x}_{2c}$ perturbations generate negative $\overline{v^2} - \overline{w^2}$ while $\mathbf{x}_{1c}$ perturbations generate positive values.

To explain these envelopes, consider the profile located at $y/h = 1$, the wall bisector. The fluid here behaves like a channel flow (assuming small span-wise gradients), such that the eigenvectors v_{i1} and v_{i3} of the a_{ij} tensor are aligned at $\pm 45^\circ$ and v_{i2} is aligned in the span-wise direction. The $\mathbf{x}_{1c}$ perturbations redistribute the normal stress so that it is all aligned along v_{i1} . Because of the 45° alignment the stress is shared between $\overline{u^2}$ and $\overline{v^2}$, while the span-wise stress $\overline{w^2}$ goes to zero, causing $\overline{v^2} > \overline{w^2}$. For $\mathbf{x}_{2c}$ perturbations, the normal stress is distributed equally between the v_{i1}

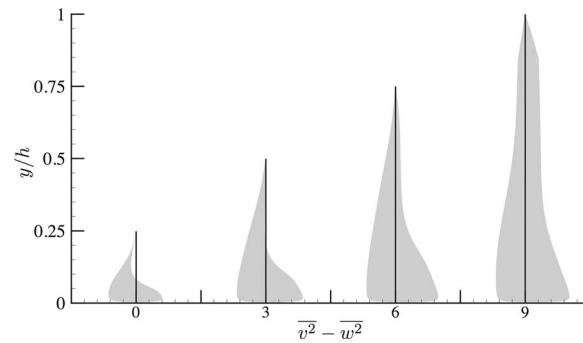


FIG. 6. Square duct profiles of $\overline{v'^2} - \overline{w'^2}$ at different span-wise positions (offset by a value of 3): from left to right are $y/h = 0.25, 0.5, 0.75$, and 1.0 . The gray regions depict the envelope between profiles from $\mathbf{x}_{1c}$ (positive values) and $\mathbf{x}_{2c}$ (negative values) perturbations with $\Delta_B = 0.5$. The $\overline{v'^2} = \overline{w'^2}$ reference (solid line) is representative of both the base-model and $\mathbf{x}_{3c}$ perturbation results.

and v_{i2} directions. The stress along v_{i1} is shared between $\overline{u'^2}$ and $\overline{v'^2}$ while the stress along v_{i2} is identically $\overline{w'^2}$, thus $\overline{v'^2} < \overline{w'^2}$.

The UQ methodology is able to introduce perturbations that indeed create secondary flows with the intensity of the streamwise vorticity directly proportional to the perturbation amplitude. These results are particularly significant because no prior knowledge of this problem was used in the design of the perturbations, i.e., the Λ^* procedure was not developed specifically to generate stream-wise vorticity. Similar to the results of the plane channel, this uncertainty injection produces envelopes around the base-model prediction, despite the fact that the mechanism and particular perturbation directions which produce the envelopes are different.

To close this section, let us consider how this could be used in practice. Consider the example of a scramjet combustor, which has a relatively complex flow field where there is a potential for secondary flows. Without any knowledge of whether there truly are any secondary flows in the scramjet, the UQ methodology could be used to perturb the RANS model such that secondary flows are produced, and the effect of this on the quantity-of-interest (e.g., the thrust) could be assessed. If this effect is large, then one should question the reliability of the eddy-viscosity predictions for this problem; however, if it is small, then there is no reason to distrust the RANS despite knowing that it fails to predict secondary flows.

C. Delery transonic bump

We next investigate a more physically complex case where transonic flow, shock-boundary layer interaction, streamline curvature, and separation are present: Delery's Case C.²⁰ Calculations are performed on a mesh containing 280×100 elements in the stream-wise and wall-normal directions, respectively, and the near-wall grid is designed to give y^+ of order 1 for the first line of cells. The computational domain extends upstream of the start of the bump so that a uniform inflow condition is used and boundary layers develop along the walls. The extent upstream is specified so the boundary layer at the start of the bump coincides with the experimental boundary layer height at this location. The walls are assumed to be adiabatic.

The distribution of Mach number over the bump is reported in Fig. 7. The inflow conditions correspond to $M = 0.6$ and the flow becomes supersonic (up to $M = 1.4$) over the bump, forming a curved shock bridging the entire channel. This in turn leads to a strong interaction between the shock and the bottom wall boundary layer, creating a distinctive lambda shock structure and a separation bubble at the back of the bump.

This case is often used in turbulence model assessment, typically observing the models ability or inability to predict the correct shock structure. In Fig. 8, we compare predictions of wall pressure to reference data,²¹ as well as wall shear stress for all $\mathbf{x}^{(i)}$ with $\Delta_B = 0.5$. Though reference data are

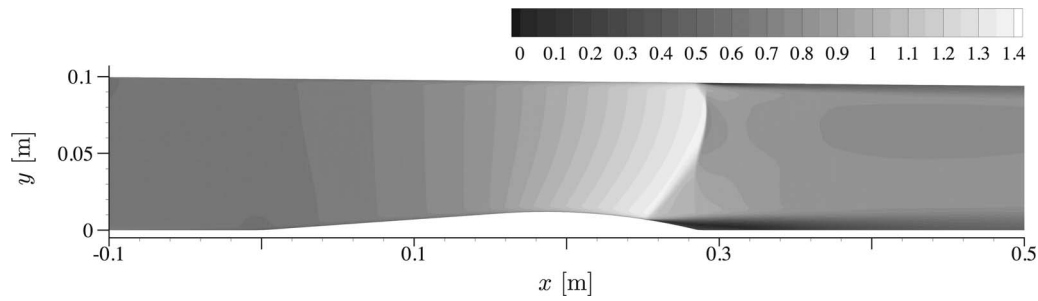


FIG. 7. Delery Case C geometry showing Mach number contours generated by the base-model. The transonic flow creates a separation bubble downstream of the bump which interacts with the lambda shock at the base of the bump.

unavailable for the shear stress, the results are significant in that they demonstrate the uncertainty injection's influence on different quantities-of-interest.

There are two important observations: first, the wall pressure is clearly much less sensitive to the perturbations than the wall shear stress. The perturbed pressure profiles only deviate from the base-model downstream of the separation point. This is due to the fact that the wall pressure is determined by the free-stream, which is modified only minutely by the perturbations (due to boundary layer growth rates). The important conclusion is that the simulation's response to the injection of uncertainty depends on which quantities-of-interest are being observed. Second, the enveloping behavior is again observed, similar to the analysis of plane shear flow. With the exception of the shock wave and some points in the separation bubble, the flow in much of the domain is similar to a parallel shear flow; presumably the same mechanisms which caused the envelopes for the channel are at work here. The sensitivity of these envelopes to Δ_B is shown in Fig. 9, where again the region between $\mathbf{x}_{1c}$ and $\mathbf{x}_{3c}$ profiles is filled in.

The injection of uncertainty affects more than just the wall quantities. In Fig. 10 are streamlines showing the size of the separation bubble formed at the back of the bump geometry. The size of this bubble is a different quantity-of-interest where enveloping of the base-model occurs.

IV. RESULTS: NON-UNIFORM Δ_B

In Sec. II, we stated that the first component in this UQ framework is identification of regions in the flow where the modeling assumptions are locally inappropriate or questionable. In Sec. III, we skipped this step and injected uncertainty uniformly throughout the computational domain.

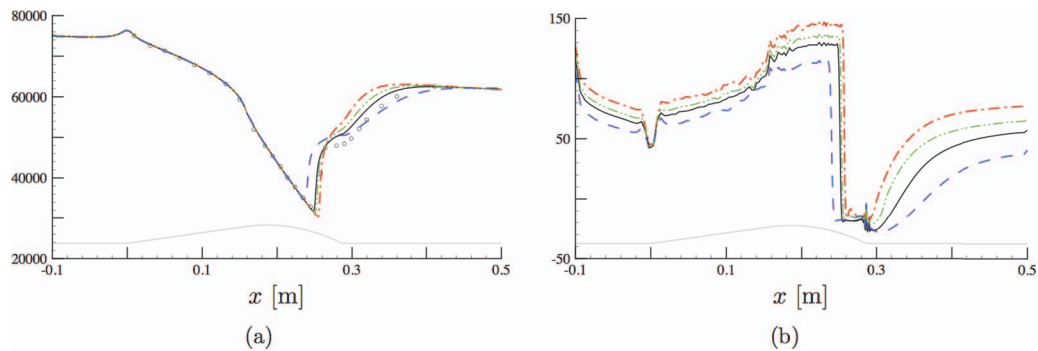


FIG. 8. Delery Case C profiles of wall pressure (a) and wall shear stress (b) along the bottom wall (geometry is depicted by the gray line). The base-model solution (solid line) is enveloped by the eigenvalue perturbations. In (b), $\mathbf{x}_{3c}$ perturbations (dashed) under-predict the magnitude of τ_w while $\mathbf{x}_{2c}$ (dashed-dotted-dotted) and $\mathbf{x}_{1c}$ (dashed-dotted) over-predict relative to the base-model. The P_w profiles in (a) behave slightly differently in that the upper and lower biases of the perturbations switch at the base of the bump geometry. Downstream they bias in the same direction as in (b), however upstream they are inverted (though still envelope the base-model). Experimentally measured P_w (circles²¹) is provided for reference in (a).

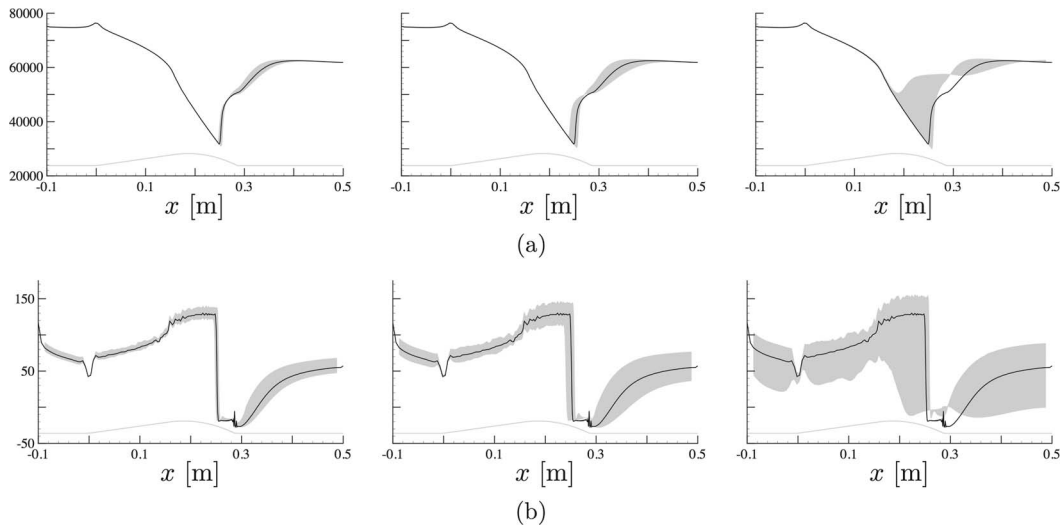


FIG. 9. Delery Case C profiles of P_w (a) and τ_w (b) along the bottom wall (geometry depicted by gray line). Displayed are envelopes (gray region between $\mathbf{x}_{3c}$ and $\mathbf{x}_{1c}$ profiles) for eigenvalue perturbations with different values of Δ_B . From left to right are $\Delta_B = 0.25, 0.5$, and 1.0 perturbation results, displaying a clear relationship between the amplitude of perturbation and the size of the envelope. The base-model solution (solid line) is provided for reference.

Uniform injection throughout the domain clearly risks over-estimating the uncertainty by perturbing in regions where we really trust the RANS model, but it also might lead to under-estimating the effects of the perturbations by ignoring their gradient. Once an actual “marker function” is used to identify where in the domain uncertainty should be injected, the perturbation magnitude Δ_B will vary in space. And, in principle, the target state toward which we are perturbing could vary in space too.

The objective of this section is to assess how the uncertainty-injection process is affected by a non-uniform Δ_B . Note that we are not attempting to develop and validate a reasonable and universal “marker function” here, but instead focus on how the injection process is affected by such a marker.

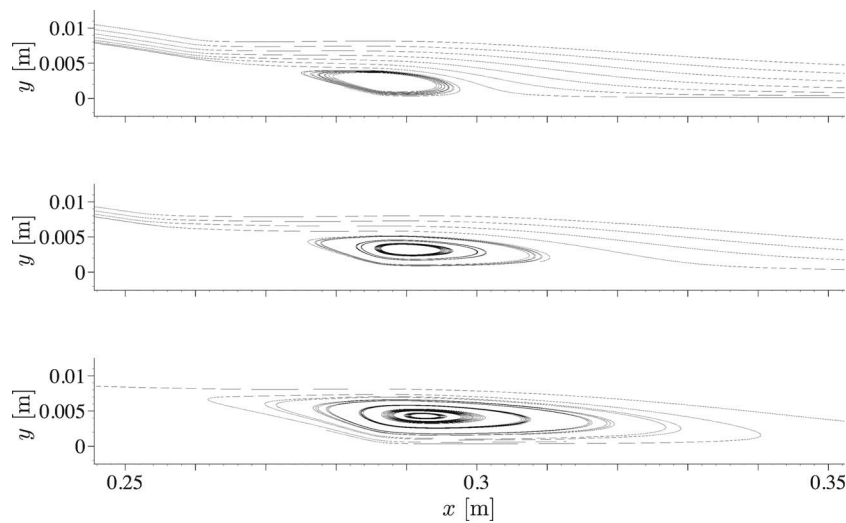


FIG. 10. Delery Case C streamlines showing the size of the separation bubble downstream of the bump for the base-model (b), and $\mathbf{x}_{1c}$ (a) and $\mathbf{x}_{3c}$ (c) perturbations where $\Delta_B = 0.5$. The eigenvalue perturbations influence the flow beyond just wall quantities and are able to envelope the size of the separation phenomenon.

It is the divergence of the Reynolds stress ($-\partial R_{ij}/\partial x_j$) that enters the mean momentum equations. For the canonical case of plane shear flow, this can be decomposed as

$$\frac{\partial}{\partial y} R_{12}^*|_{\mathbf{x}_{1c}} = \overbrace{(1 - \Delta_B) \frac{\partial R_{12}}{\partial y}}^{\text{Part 1}} - \overbrace{(R_{12} + k\Psi) \frac{\partial \Delta_B}{\partial y}}^{\text{Part 2}} - \overbrace{\Delta_B \Psi \frac{\partial k}{\partial y}}^{\text{Part 3}} - \overbrace{\Delta_B k \dot{\Psi} \frac{\partial \theta}{\partial y}}^{\text{Part 4}}, \quad (11a)$$

$$\frac{\partial}{\partial y} R_{12}^*|_{\mathbf{x}_{2c}} = (1 - \Delta_B) \frac{\partial R_{12}}{\partial y} - \left(R_{12} + \frac{2}{3} k \Psi \right) \frac{\partial \Delta_B}{\partial y} - \frac{2}{3} \Delta_B \Psi \frac{\partial k}{\partial y} - \frac{2}{3} \Delta_B k \dot{\Psi} \frac{\partial \theta}{\partial y}, \quad (11b)$$

$$\frac{\partial}{\partial y} R_{12}^*|_{\mathbf{x}_{3c}} = (1 - \Delta_B) \frac{\partial R_{12}}{\partial y} - R_{12} \frac{\partial \Delta_B}{\partial y}, \quad (11c)$$

where $\Psi = \sin 2\theta$ and $\dot{\Psi} = 2 \cos 2\theta$. Recall that θ is angle of the eigenvectors of a_{ij} , thus $\theta = 45^\circ$ for the present eddy-viscosity base-model implying that Part 4 is zero here.

The second term (“Part 2”) is proportional to the gradient of Δ_B , suggesting that a sufficiently rapidly varying perturbation magnitude could completely dominate everything else. This, however, is not reasonable. The perturbations attempt to model (in some sense) the systematic bias between the RANS predictions and the truth. Therefore, the perturbations should have a spatial variability that is commensurate with the variability of the RANS bias. For example, in channel flow, one should expect the RANS bias to vary between the buffer layer and the log-layer, over about the same distance that the turbulence itself adjusts between these layers. Thus, the perturbation magnitude Δ_B should be expected to vary in general, but not abruptly.

In the following, we revisit the three test problems and consider different types of non-uniform distributions of Δ_B .

A. Plane channel flow

The availability of DNS data for the plane channel flow problem implies that the terms in Eq. (11) can be estimated using “truth data.” Specifically, the key question is how large the Part 2 contribution (gradient of Δ_B) is compared to the other contributions. This is pursued first, and then some more general distributions of Δ_B are considered.

1. A priori analysis of Δ_B variation

For each value of y^+ , the locations in the barycentric map for the DNS data and the RANS base-line model are $\mathbf{x}_{\text{DNS}}$ and $\mathbf{x}_{\text{RANS}}$, respectively. For each target state $\mathbf{x}^{(t)}$, there exists a value of Δ_B that minimizes the distance (in the barycentric map) between $\mathbf{x}_{\text{DNS}}$ and the perturbed state of the RANS; this minimum distance is labeled γ . Perturbations toward $\mathbf{x}_{1c}$ generate the minimum γ at every y^+ . This minimum γ is shown in Fig. 11(a), along with the “optimum” Δ_B profile. The point in using this particular Δ_B profile is that it has a spatial variation directly related to the spatial variation of the RANS errors. This can then be used to evaluate the different terms in Eq. (11), the results of which are shown in Fig. 11(b). Note that R_{12} is neglected in Part 2; since this is negative, this is a conservative estimate that over-estimates the contribution of Part 2.

It is clear that Part 3 is the largest, with Part 2 being comparable only at the edge of the buffer layer around $y^+ \approx 40$ where the “optimum” Δ_B was found to vary most rapidly. In terms of integrated contributions, Part 3 is clearly dominant. This suggests that the simplified analysis in Sec. IIE (which neglects the gradient of Δ_B) is a valid way to gain understanding into the effects of the perturbations.

2. Candidate Δ_B distributions

We next consider several different distributions of the perturbation magnitude Δ_B for the purpose of seeing how its spatial variability affects the results. The different distributions are not intended to be practically relevant in describing model errors, as they are rather flow-specific.

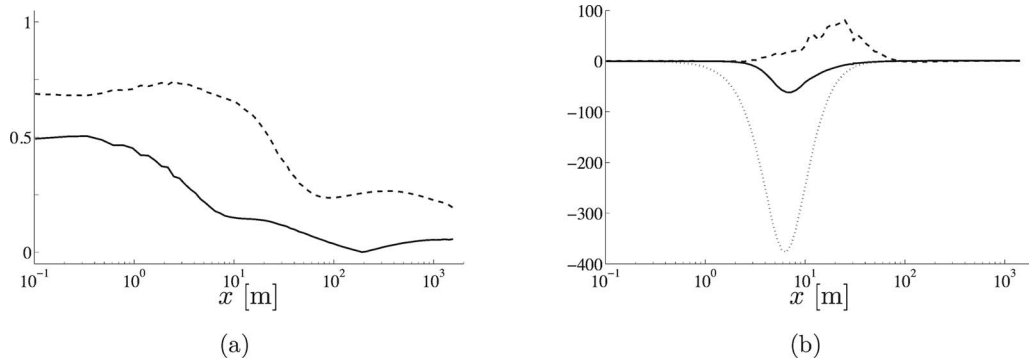


FIG. 11. Channel flow profiles of x_{1c} perturbation where an optimal Δ_B (which minimizes the RANS-DNS discrepancy) is prescribed are shown. In (a), the discrepancy distance γ (solid) and optimal Δ_B (dashed) are shown. In (b), the relative contributions to the momentum source term are shown for Part 1 (solid), Part 2 (dashed), and Part 3 (dotted).

a. Error model. The discrepancy in the turbulent kinetic energy k between the base-line RANS model and the DNS results¹⁸ is shown in Fig. 12. The discrepancy is largest in the buffer layer, while the agreement is generally better in the viscous sublayer and the log-layer. Thus, one possible type of spatial dependency mimics this discrepancy: an error model. The marker is defined as a combination of two log-normal distributions where the observed error (see Fig. 12(b)) is represented notionally. This distribution is scaled such that $\Delta_{B, \max} = 1$, shown in Fig. 14(a).

b. Random. To generate distributions of Δ_B that vary more rapidly, linear combinations of two log-normal distributions where the parameters (mean and variance) are chosen randomly are considered. These distributions are in the form

$$\Delta_B(y) \propto \ln \mathcal{N}(y; \mu_1, \sigma_1^2) + \ln \mathcal{N}(y; \mu_2, \sigma_2^2),$$

where the amplitude is scaled such that $\Delta_{B, \max} = 1$. The parameters are chosen randomly from $\mu_1, \mu_2, \sigma_1, \sigma_2 \in [0, 1]$. Two samples are shown in Fig. 14(b).

c. Wall-distance based model. In the current work, uncertainty is injected by perturbing the anisotropy eigenvalues, thus a more logical approach for the marker would leverage the discrepancy in those eigenvalues (λ_i), displayed in Fig. 13. Each of these three profiles could be used to construct a separate error model, however, we leverage the shared trend that the discrepancy is larger for $y^+ < 100$. Combined with our understanding that eddy-viscosity models do not properly characterize

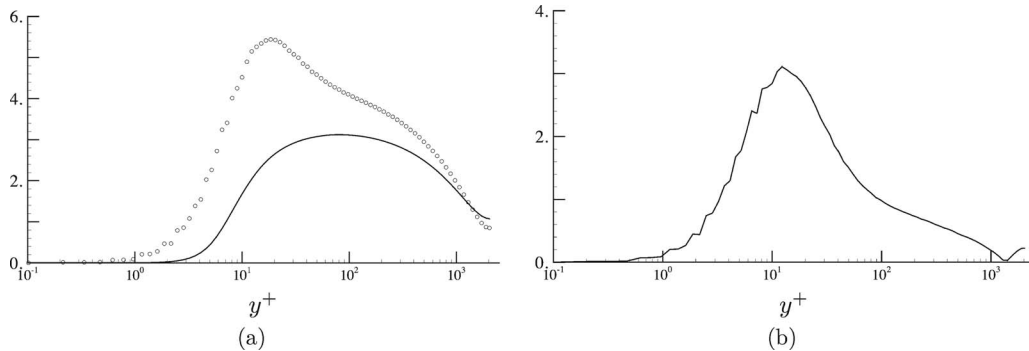


FIG. 12. Channel flow profiles of turbulence kinetic energy versus distance from the wall (a) for DNS ($\circ$) and the base-model (solid line). The discrepancy between the RANS base-model and DNS ($|k_{\text{DNS}} - k_{\text{RANS}}|$) is displayed in (b).

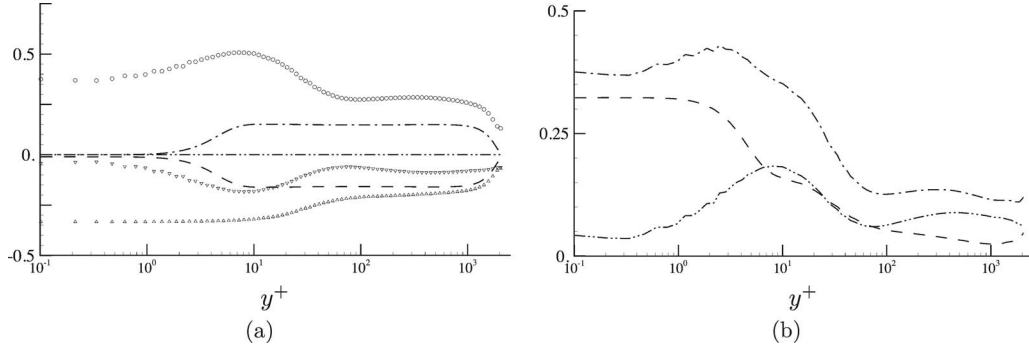


FIG. 13. Channel flow turbulence anisotropy eigenvalues versus distance from the wall (a) for DNS (λ_1 : $\circ$, λ_2 : ∇ , λ_3 : $\triangle$) and the base-model (λ_1 : dashed-dotted, λ_2 : dashed-dotted-dotted, λ_3 : dashed). The base-model discrepancy in these eigenvalues $|\lambda_{I,DNS} - \lambda_{I,RANS}|$ is displayed in (b).

the near wall componentiality of turbulence,²² we define a marker based on wall-distance

$$\Delta_B = \Delta_{B,\max} \frac{1}{2} \left[1 - \tanh \left(C_{\tanh} \frac{(d_w - d_w^{\text{TH}})}{h} \right) \right], \quad (12)$$

where d_w is the current element's wall-distance and $C_{\tanh} = 50$ (controls the steepness of the transition between left and right states). At a user specified threshold wall-distance d_w^{TH} , this function smoothly switches Δ_B between two uniform distributions of $\Delta_{B,\max}$ and zero. In the following results $d_w^{\text{TH}} = 0.1h$ and $\Delta_{B,\max} = 0.5$, shown in Fig. 14(c). Note that the marker is simply the distance from the wall, where Eq. (12) describes the relationship between the marker and Δ_B . This marker is easily computed and applicable for any flow geometry, in contrast to the error modeled distribution which is unique to the channel flow.

3. Results

We first discuss results obtained using the error model and stochastic markers. The profiles for $\mathbf{x}_{1c}$ and $\mathbf{x}_{3c}$ perturbations are plotted relative to the base-model prediction in Fig. 15. Both of these methods normalized the profiles such that $\Delta_{B,\max} = 1$ so their results are overlaid with the envelope generated by the uniform $\Delta_B = 1$ simulations. There are three important observations from these results. The first is that while in general enveloping behavior is maintained, there are several instances where the base-model solution is not encapsulated. These regions appear correlated to large gradients of Δ_B . Second, in several locations the perturbations which produce the upper and lower bound switch, e.g., the buffer layer for the second stochastic profile. The third observation is that in several locations the envelope based on uniform Δ_B is exceeded in magnitude. All three of these observations are present when using the wall-distance marker as well. These results, shown relative to the $\Delta_B = 0.5$ envelopes, are in Fig. 16.

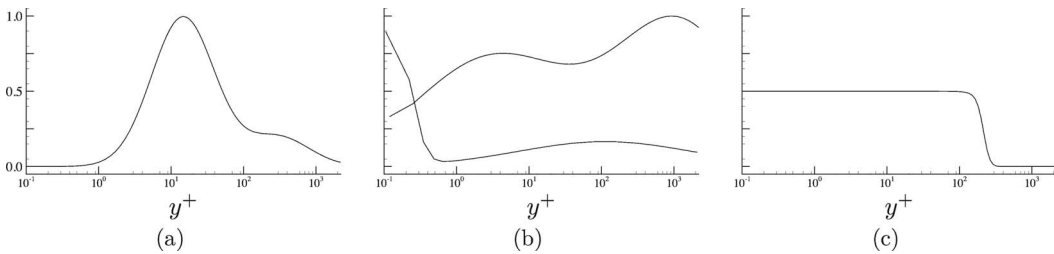


FIG. 14. Channel flow profiles of Δ_B versus distance from the wall for different marker generation philosophies. The distributions correspond to an error model (a), randomly generated (b), and a wall-distance based marker (c).

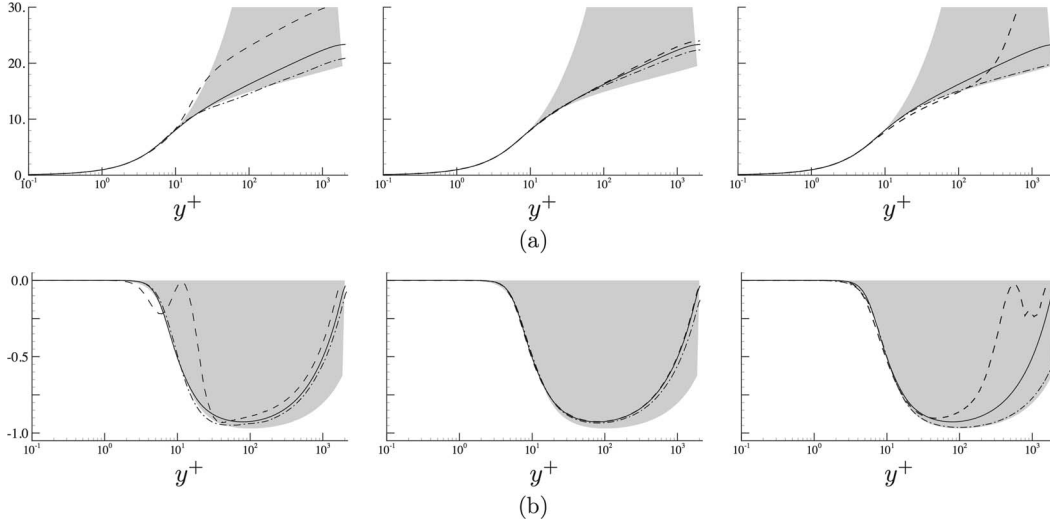


FIG. 15. Channel flow profiles of u^+ (a) and $\overline{u'v'}/u_\tau^2$ (b) versus distance from the wall. The $\mathbf{x}_{1c}$ (dashed-dotted) and $\mathbf{x}_{3c}$ (dashed) perturbation profiles for the error modeled marker (left) and stochastic marker (middle and right) are shown, with the base-model (solid) provided for reference. The gray regions are the envelopes generated from uniform $\Delta_B = 1.0$ perturbations, displayed because in these non-uniform distributions $\Delta_{B, \max} = 1$.

Clearly, Part 2 of Eq. (11) is causing a large change in the momentum source term. And, when the magnitude of $\partial_2 \Delta_B$ is sufficiently large, the plane shear flow analysis no longer applies throughout the domain.

These results must be balanced against the results of the *a priori* study, in which it was found that the perturbation magnitude Δ_B as deduced from DNS data varies less rapidly in space than the error models considered here.

B. Square duct flow

The wall-distance marker is used for the square duct with $d_w^{TH} = 0.2h$. Section IV A provided a discussion of how the spatial distribution of Δ_B influences the solution, here we simply show the result when applied to this flow. The wall distance marker is an appropriate choice for the same reasons described for the channel, namely, that RANS models poorly characterize turbulent structures near walls within attached boundary layers.

The velocity vectors in Fig. 17 confirm that secondary flow is introduced even with localized injection of uncertainty, however, the recirculating vortices are restricted to the marker region.

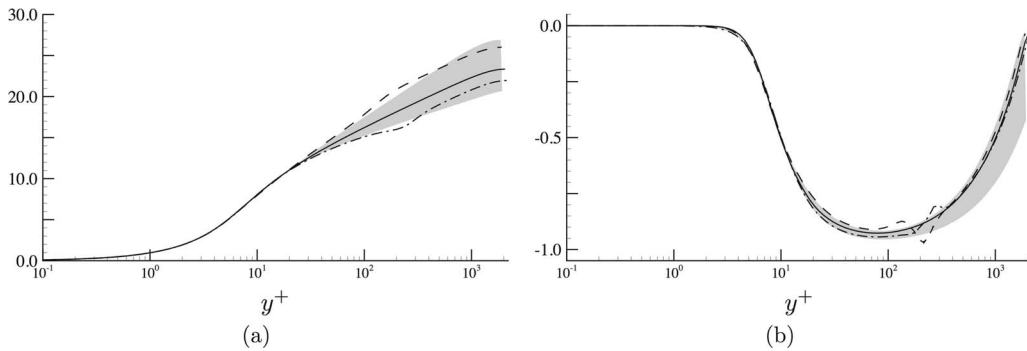


FIG. 16. Channel flow profiles of u^+ (a) and $\overline{u'v'}/u_\tau^2$ (b) versus distance from the wall. The $\mathbf{x}_{1c}$ (dashed-dotted) and $\mathbf{x}_{3c}$ (dashed) perturbation profiles for the wall-distance marker are shown, with the base-model (solid) provided for reference. The gray regions are the envelopes generated from uniform $\Delta_B = 0.5$ perturbations, displayed because in these non-uniform distributions $\Delta_{B, \max} = 0.5$.

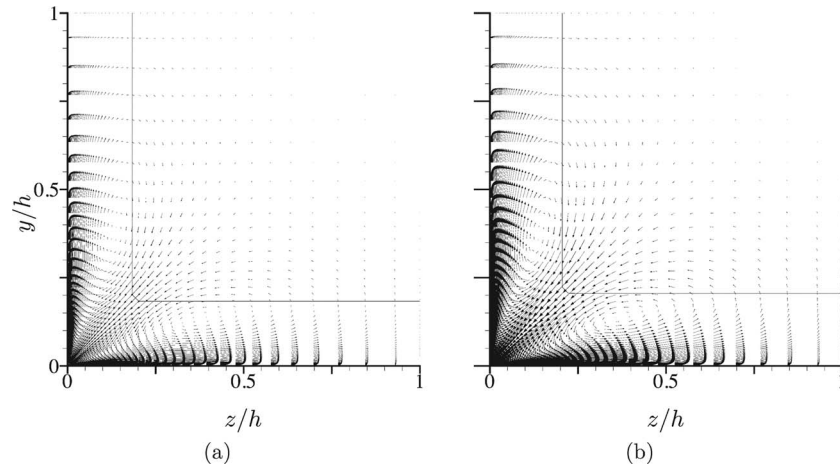


FIG. 17. Square duct results for $\mathbf{x}_{2c}$ eigenvalue perturbations using a wall-distance marker where $d_w^{TH} = 0.2h$ (indicated by the contour line). In-plane velocity vectors are used to visualize the presence of secondary flows. A quadrant of the duct at a $y - z$ slice at $x = 190/h$ is displayed. Secondary flows are generated, though restricted primarily to the marker region, and the strength of the secondary flows are related to the amplitude of perturbation where $\Delta_B = 0.25$ (a) and 0.5 (b).

Fig. 18 shows the $\overline{v^2} - \overline{w^2}$ profiles using this marker relative to the uniform $\Delta_B = 0.5$ envelopes. With this choice of marker, the enveloping behavior is still observed, where deviation from the base-model profile occurs only where the marker is active (greater than zero). The results presented indeed illustrate the effectiveness of the proposed approach in determining solutions that bound the baseline model.

C. Delery transonic bump

The distributions of the perturbation magnitude Δ_B used above are not universally applicable; they were used only to assess the effect of spatial variability in Δ_B . For the final test case, however, let us introduce a more generally applicable marker function, as an indication for how the complete UQ machinery could operate.

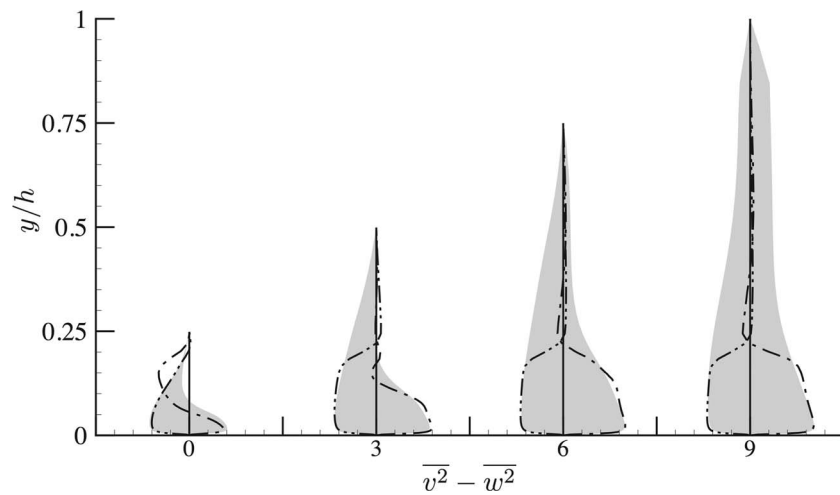


FIG. 18. Square duct profiles of $\overline{v^2} - \overline{w^2}$ at different span-wise positions (offset by a value of 3): from left to right are $y/h = 0.25, 0.5, 0.75$, and 1.0 . Displayed are profiles of $\mathbf{x}_{1c}$ (dashed-dotted) and $\mathbf{x}_{2c}$ (dashed-dotted-dotted) perturbations for the wall-distance marker with $d_w^{TH} = 0.2h$. The $\overline{v^2} - \overline{w^2}$ reference (solid line) is representative of both the base-model and $\mathbf{x}_{3c}$ perturbation results. The gray regions are the envelopes generated from uniform $\Delta_B = 0.5$ perturbations, displayed because in these marker-based results $\Delta_{B, \max} = 0.5$.

Most validation cases (which we define here as *the available evidence* on how well a RANS closure works) are for parallel or nearly parallel shear flows: e.g., plane and round jets, mixing layers, channel and boundary layer flows. Good turbulence models make useful and reasonable predictions for these flows. We construct a marker which attempts to identify deviations from parallel shear flow.

To identify these regions in a generic configuration, we consider the local velocity gradients along the streamlines. Consider a local velocity vector u_i and the unit vector along the streamline $s_i = u_i / \sqrt{u_n u_n}$. The vector

$$g_j = s_i \frac{\partial u_i}{\partial x_j} \quad (13)$$

is then the gradient of the streamline-aligned velocity. The ratio

$$f = \frac{|g_j s_j|}{\sqrt{g_n g_n}} \quad (14)$$

is then the cosine of the angle between the direction of the streamline and the gradient of the velocity projected onto the streamline. For perfectly parallel shear flows, with no variation in the stream-wise direction (assuming divergence-free velocity), f is exactly zero. Thus, f is a local measure of deviation from parallel shear flow.

We cannot use f directly as the marker, however, for a few reasons: first, it becomes undefined (k goes to zero) in bulk or free-stream flows. Second, in regions of vanishing turbulence, errors in R_{ij} have no impact. While this also implies that we could freely perturb R_{ij} in those regions, it makes sense not to do so to avoid numerical issues. Therefore, we define the actual marker as

$$m = f \frac{k}{u_n u_n}, \quad (15)$$

such that f is scaled by local turbulence intensity, approximating the relative importance of the turbulent stresses over the convection. Based on this marker function, the perturbation magnitude is then taken as

$$\Delta_B = \Delta_{B,\max} \begin{cases} 0.5 - 0.5 \cos(\pi m / m_c), & m < m_c \\ 1, & m \geq m_c \end{cases}, \quad (16)$$

where $\Delta_{B,\max} = 0.5$ and $m_c = 0.001$ is a parameter controlling the magnitude of m at which perturbations should occur.

A thorough validation of this marker would imply an assessment of how well it identifies regions of RANS errors in a wide range of different types of flows. This is left for future work, and we instead merely use the marker here to assess its effects on the Delery test case.

The perturbed results are shown in Fig. 19. The $\mathbf{x}_{1c}$ and $\mathbf{x}_{3c}$ perturbations are displayed along with the envelope from the uniform $\Delta_B = 0.5$. Using this marker the pressure envelope is essentially

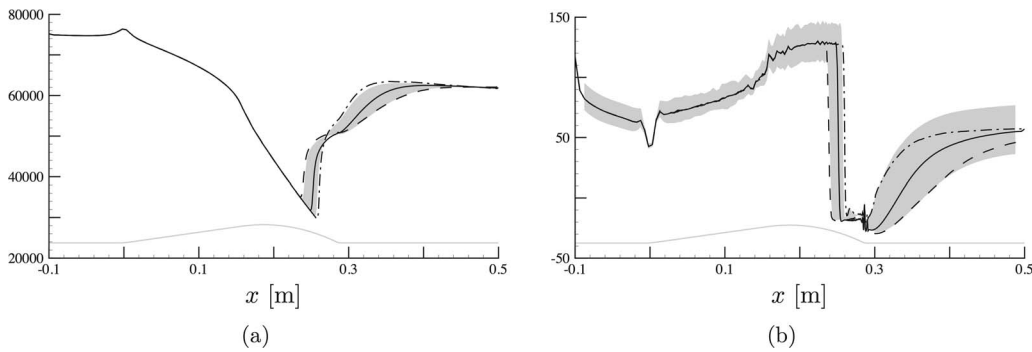


FIG. 19. Delery Case C profiles of P_w (a) and τ_w (b) along the bottom wall (geometry depicted by gray line). Profiles of $\mathbf{x}_{3c}$ (dashed) and $\mathbf{x}_{1c}$ (dashed-dotted) perturbations using the marker f and $\Delta_{B,\max} = 0.5$ are shown. The base-model solution (solid line) and the envelopes (gray region) generated from uniform $\Delta_B = 0.5$ perturbations are provided for reference.

unchanged. There are two regions where the shear stress profiles show significant deviation from the uniformly distributed result: the intake upstream of the bump geometry ($x < 0$) and at the end of the domain ($x > 0.4$). Upstream this is simply a channel flow exhibiting parallel shear flow, so the marker is inactive, i.e., $\Delta_B \approx 0$. The same argument is true for the channel downstream of the bump, as the flow returns to parallel shear conditions the perturbed profiles return toward the base-model prediction.

V. SUMMARY

The problem of estimating the effects of model-form uncertainties in RANS models is tackled. The guiding philosophy is to seek reasonable estimates of this uncertainty rather than mathematically rigorous bounds. A specific objective of this work is to develop a framework for model-form uncertainty estimation that goes beyond parametric uncertainties; specifically, a framework that includes uncertainty in the underlying eddy-viscosity hypothesis is sought.

The proposed framework has essentially two components. The first is to identify spatial regions in the flow where the RANS model is plausibly erroneous. It would be much too pessimistic to assume that a well-validated RANS model is wrong *everywhere*. Instead, it is much more realistic and practical to assume that the model is correct in those regions where the flow is similar to what the model was designed and validated for (e.g., attached boundary layers), and that the model is erroneous in other, more complex, flow-regions. The development of such *marker* functions that identify these different regions is part of ongoing work, though some candidates were presented and demonstrated in this paper.

The second component of the overall framework, and the subject of this contribution, is the injection of uncertainty. The proposed approach to this injection problem is that it should be done directly into the Reynolds stress tensor, as opposed to uncertainty injection into the coefficient in some postulated model equations. Injecting into the Reynolds stress directly brings multiple benefits: the method becomes model-independent (applicable to many models, including non-eddy-viscosity ones), and, most importantly, errors in the eddy-viscosity hypothesis are included in the framework. The injection of uncertainty is done by decomposing the stress tensor into its size, shape, and orientation; i.e., the turbulent kinetic energy, the stress anisotropy, and the orientation of the stress eigenvectors. The main advantage of this decomposition is that it makes possible physical interpretation of different types of perturbations. The focus of this paper is on perturbations to the shape or anisotropy, these perturbations are introduced through the barycentric map.

Results are presented for incompressible flow in a turbulent channel and duct and for the transonic flow over a curved bump. Of particular importance is the study of the duct flow in which secondary flows are generated solely by the turbulence anisotropy and therefore not predicted by the original eddy-viscosity closure. The proposed perturbation framework is capable of generating secondary recirculation. The secondary flow is physically realistic when perturbing toward the two-component limit, which is the correct physical limit when approaching a solid wall. Interestingly, a completely unphysical “reverse” secondary flow is generated when perturbing toward the one-component limit.

ACKNOWLEDGMENTS

This research was supported by the (U.S.) Department of Energy (DOE) [National Nuclear Security Administration] under Award No. NA28614.

¹ A. Raftery, T. Gneiting, F. Balabdaoui, and M. Polakowski, “Using Bayesian model averaging to calibrate forecast ensembles,” *Mon. Weather Rev.* **133**, 1155 (2005).

² D. Wilcox, *Turbulence Modeling for CFD* (DCW Industries, Inc., La Cañada, CA, 1994).

³ P. Platteeuw, G. Loeven, and H. Bijl, “Uncertainty quantification applied to the k - ϵ model of turbulence using the probabilistic collocation method,” AIAA Paper No. 2008-2150, 2008.

⁴ M. C. Dunn, B. Shotorban, and A. Frendi, “Uncertainty quantification of turbulence model coefficients via Latin hypercube sampling method,” *J. Fluids Eng.* **133**, 041402 (2011).

- ⁵ T. Oliver and R. Moser, "Bayesian uncertainty quantification applied to RANS turbulence models," *J. Phys.: Conf. Ser.* **318**, 042032 (2011).
- ⁶ S. Cheung, T. Oliver, E. Prudencio, S. Prudhomme, and R. Moser, "Bayesian uncertainty analysis with applications to turbulence modeling," *Reliab. Eng. Syst. Saf.* **96**, 1137 (2011).
- ⁷ G. Iaccarino, R. Pečnik, J. Glimm, and D. Sharp, "A QMU approach for characterizing the operability limits of air-breathing hypersonic vehicles," *Reliab. Eng. Syst. Saf.* **96**(9), 1150 (2011).
- ⁸ G. Iaccarino, D. Sharp, and J. Glimm, "Quantification of margins and uncertainties using multiple gates and conditional probabilities," *Reliab. Eng. Syst. Saf.* **114**, 99 (2013).
- ⁹ P. Bradshaw, "Turbulent secondary flows," *Annu. Rev. Fluid Mech.* **19**, 53 (1987).
- ¹⁰ L. Jones, P. Gaskell, H. Thompson, X. Gu, and D. Emerson, "Anisotropic, isothermal, turbulent swirling flow in a complex combustor geometry," *Int. J. Numer. Methods Fluids* **47**, 1053 (2005).
- ¹¹ A. Bhattacharya, S. Kassinos, and R. Moser, "Representing anisotropy of two-point second-order turbulence velocity correlations using structure tensors," *Phys. Fluids* **20**, 101502 (2008).
- ¹² C. Górlé and G. Iaccarino, "A framework for epistemic uncertainty quantification of turbulent scalar flux models for Reynolds-averaged Navier-Stokes simulations," *Phys. Fluids* **25**, 055105 (2013).
- ¹³ L. Terentiev, "The turbulence closure model based on linear anisotropy invariant analysis," Ph.D. thesis, Universität Erlangen-Nürnberg, 2006.
- ¹⁴ C. Speziale, R. Abid, and P. Durbin, "On the realizability of Reynolds stress turbulence closures," *J. Sci. Comput.* **9**, 369 (1994).
- ¹⁵ S. Banerjee, R. Krahle, F. Durst, and C. Zenger, "Presentation of anisotropy properties of turbulence, invariants versus eigenvalue approaches," *J. Turbul.* **8**, N32 (2007).
- ¹⁶ F. Menter, "Two-equation eddy-viscosity turbulence models for engineering applications," *AIAA J.* **32**(8), 1598 (1994).
- ¹⁷ R. Pečnik, V. Terrapon, F. Ham, G. Iaccarino, and H. Pitsch, "Reynolds-averaged Navier-Stokes simulations of the HyShot II scramjet," *AIAA J.* **50**(8), 1717 (2012).
- ¹⁸ S. Hoyas and J. Jimenez, "Scaling of the velocity fluctuations in turbulent channels up to $Re_\tau = 2003$," *Phys. Fluids* **18**, 011702 (2006).
- ¹⁹ R. Pečnik and G. Iaccarino, "Predictions of turbulent secondary flows using the $v^2 - f$ model," AIAA Paper No. 2008-3852, 2008.
- ²⁰ J. Delery, "Investigation of strong shock turbulent boundary layer interaction in 2-D transonic flows with emphasis on turbulence phenomena," AIAA Paper No. 1981-1245, 1981.
- ²¹ A. Hadjadj and A. Kudryavtsev, "Computation and flow visualization in high-speed aerodynamics," *J. Turbul.* **6**, N16 (2005).
- ²² B. B. Launder and N. D. Sandham, *Closure Strategies for Turbulent and Transitional Flows* (Cambridge University Press, Cambridge, 2002).